

Unit ID: 67349731ef47416e2ea93610

unit 1

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sample data

Gallery Images

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PDF Documents

PDF Document 1: 1731499824532-arithmetic.pdf

This PDF document is included in the unit materials.

Videos

Video file: 1731499824540-5538137-hd_1920_1080_25fps.mp4

YouTube Links

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1. Addition (+)

Definition:

Addition involves combining two or more quantities to find their total.

Example:

- $5+3=8$ $5 + 3 = 8$

Properties of Addition:

- **Commutative Property:** $a+b=b+a$ $a + b = b + a$
- **Associative Property:** $(a+b)+c=a+(b+c)$ $(a + b) + c = a + (b + c)$
- **Identity Property:** $a+0=a$ $a + 0 = a$

Applications of Addition:

- **Shopping:** Adding prices of items to find the total bill.
 - **Finance:** Calculating total income or expenses over a period.
 - **Measurements:** Adding distances or weights to find cumulative values.
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2. Subtraction (-)

Definition:

Subtraction involves finding the difference between two numbers, representing how much one number exceeds the other.

Example:

- $8-3=5$ $8 - 3 = 5$

Properties of Subtraction:

- **Non-Commutative:** $a-b \neq b-a$ $a - b \neq b - a$
- **Identity Property:** $a-0=a$ $a - 0 = a$

Applications of Subtraction:

- **Banking:** Calculating how much money remains after a withdrawal.
 - **Inventory:** Finding how many items are left after some are sold.
 - **Travel:** Calculating how much distance remains in a journey.
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3. Multiplication (×)

Definition:

Multiplication is repeated addition, where a number is added to itself a specific number of times.

Example:

- $3 \times 4 = 3 + 3 + 3 + 3 = 12$ $12 \times 4 = 3 + 3 + 3 + 3 = 12$ $3 \times 4 = 3 + 3 + 3 + 3 = 12$

Properties of Multiplication:

- **Commutative Property:** $a \times b = b \times a$ $a \times b = b \times a$ $a \times b = b \times a$
- **Associative Property:** $(a \times b) \times c = a \times (b \times c)$ $(a \times b) \times c = a \times (b \times c)$ $(a \times b) \times c = a \times (b \times c)$
- **Distributive Property:** $a \times (b + c) = a \times b + a \times c$ $a \times (b + c) = a \times b + a \times c$ $a \times (b + c) = a \times b + a \times c$
- **Identity Property:** $a \times 1 = a$ $a \times 1 = a$ $a \times 1 = a$

Applications of Multiplication:

- **Commerce:** Calculating the total price for multiple items of the same cost.
 - **Area Measurement:** Finding the area of rectangles (length $\times$ width).
 - **Data Analysis:** Scaling values, such as multiplying population growth rates.
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4. Division ($\div$)

Definition:

Division splits a number into equal parts or determines how many times one number fits into another.

Example:

- $12 \div 4 = 3$ $12 \div 4 = 3$ $12 \div 4 = 3$

Properties of Division:

- **Non-Commutative:** $a \div b \neq b \div a$ $a \div b \neq b \div a$ $a \div b \neq b \div a$
- **Identity Property:** $a \div 1 = a$ $a \div 1 = a$ $a \div 1 = a$
- **Division by Zero:** Division by 0 is undefined.

Applications of Division:

- **Sharing:** Dividing items equally among a group (e.g., distributing candies).
 - **Finance:** Calculating the per-unit cost of a bulk purchase.
 - **Speed Calculations:** Finding average speed by dividing distance by time (e.g., $\text{Speed} = \text{Distance} \div \text{Time}$).
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5. Order of Operations (BODMAS/PEMDAS)

When multiple arithmetic operations are involved in a problem, the order of operations must be followed to get the correct answer.

BODMAS Rule:

- Brackets
- Orders (exponents or powers)
- Division and **M**ultiplication (from left to right)
- Addition and **S**ubtraction (from left to right)

Example:

$$3+6\div(2\times 3)=3+6\div 6=3+1=4 \quad 3+6\div(2\times 3)=3+6\div 6=3+1=4$$

Applications of BODMAS:

- **Mathematical Expressions:** Ensuring complex calculations are performed in the correct sequence.
- **Science and Engineering:** Evaluating formulas and equations.